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MULTICHIP PACKAGES FORMED IN STABLE STRUTURE AND METHOD OF FORMING THE SAME

Abstract

A multichip package includes a substrate connected to a plurality of solder balls including a first solder ball, a first chip stacked over or on the substrate, a second chip stacked over or on the first chip and including a first overhang not supported by the first chip, and a support structure supporting the first overhang and connected to the first solder ball that provides an external electrical connection to the substrate for a voltage external to the substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority under 35 U.S.C. § 119 (a) to Korean Patent Application No. 10-2024-0029200, filed in the Korean Intellectual Property Office, on Feb. 28, 2024, which application is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure generally relates to a multichip package formed in a stable structure with respect to physical deformation and heat treatment and a method of forming the same.

2. Related Art

[0003] Recently, as the use of portable devices such as smartphones, tablets, and wearable devices has increased, multichip packages that can demonstrate powerful performance while being smaller and lighter are being widely used. The multichip package is formed by stacking and combining multiple chips in a single package, thereby optimizing performance by strengthening interconnectivity between chips and increasing cost efficiency by increasing integration.

SUMMARY

[0004] According to an embodiment, the present disclosure may provide a multichip package including a substrate connected to a plurality of solder balls including a first solder ball, a first chip stacked over the substrate, a second chip stacked over the first chip and including a first overhang by not supported by the first chip, and a support structure supporting the first overhang and connected to the first solder ball that provides an external electrical connection to the substrate for a voltage external to the substrate.

[0005] According to an embodiment, the present disclosure may provide a multichip package including a substrate connected to a plurality of solder balls including a first solder ball, a first chip stacked over the substrate, a second chip stacked over the first chip and including a first overhang not supported by the first chip, and a support structure supporting the first overhang and connected to the first solder ball that provides an external electrical connection for a device external to the substrate.

[0006] According to an embodiment, the present disclosure may provide a multichip package including a substrate connected to a plurality of solder balls including a first solder ball and a second solder ball, a first chip stacked over the substrate, a second chip stacked over the first chip and including a first overhang not supported by the first chip, a third chip stacked over the second chip and including a second overhang not supported by the second chip, a first support structure supporting the first overhang and connected to the first solder ball that provides a first external electrical connection for the substrate, and a second support structure supporting the second overhang and connected to the second solder ball that provides a second external electrical connection for the substrate.

[0007] According to an embodiment, the present disclosure may provide a multichip package including a substrate connected to a plurality of solder balls including a first solder ball and a second solder ball, a first chip stacked over the substrate, a second chip stacked over the first chip and including a first overhang not supported by the first chip, a third chip stacked over the second chip and including a second overhang not supported by the second chip, a fourth chip stacked over the third chip and including a third overhang not supported by the third chip, a first support structure supporting the first overhang and the second overhang and connected to the first solder ball that provides an external electrical connection to the substrate for a first voltage, and a second support structure supporting the third overhang and connected to the second solder ball that provides an external electrical connection to the substrate for a second voltage.

[0008] According to an embodiment, the present disclosure may provide a method of forming a multichip package. The method may include preparing a substrate connected to a plurality of solder balls, forming a support structure over the substrate, wherein the support structure is connected an external electrical connection for the substrate, stacking a first chip over the substrate, and stacking a second chip over the first chip and the support structure. The second chip may be stacked over the first chip such that an overhang not supported by the first chip is supported by the support structure.

[0009] A multichip package comprising a substrate; a first chip stacked over the substrate; a second chip stacked over the first chip, wherein the second chip is offset from the first chip in a first direction resulting in an overhang; and a support structure disposed between the overhang and the substrate and electrically connected to an external electrical connection for the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** illustrates a configuration of a multichip package according to an embodiment of the present disclosure.

[0011] FIG. **2** and FIG. **3** illustrate top and bottom views, respectively, of a plurality of chips and a support structure according to an embodiment.

[0012] FIG. **4** and FIG. **5** illustrate top and bottom views, respectively, of a plurality of chips and support structures according to an embodiment.

[0013] FIG. **6** illustrates a configuration of a multichip package according to an embodiment of the present disclosure.

[0014] FIG. **7** illustrates a configuration of a multichip package according to an embodiment of the present disclosure.

[0015] FIG. **8** illustrates a configuration of a multichip package according to an embodiment of the present disclosure.

[0016] FIG. **9** illustrates a configuration of a multichip package according to an embodiment of the present disclosure.

[0017] FIG. **10** through FIG. **13** illustrate views of a multichip package formed utilizing a method of forming a multichip package according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0018] In the following description, “an example” and “another example” may refer to an embodiment in which the configuration, method, and technical features of the disclosure are specified. The configuration, method, and technical features of the disclosure described in one embodiment may be combined with the configuration, method, and technical feature of the disclosure described in another embodiment to form another embodiment.

[0019] Terms such as “top,” “bottom,” “over,” “above,” “below,” “X direction,” “y direction,” “Z direction,” and other terms implying specific spatial relationship and/or orientation are utilized only for ease of description or reference to a drawing and are not otherwise limiting. The cross-hatching throughout the figures illustrates corresponding or similar areas between the figures rather than indicating the materials for the areas.

[0020] In the description, when an element included in an embodiment is described in singular form, the element may be interpreted to include a plurality of elements performing the same or similar functions.

[0021] Terms such as “first” and “second” are used to distinguish between various elements and do not imply size, order, priority, or importance of the elements. For example, a first element may be named as a second element in one example, and the second element may be named as a first element in another example.

[0022] When one element is identified as “connected” to another element, the elements may be connected directly or through an intervening element between the elements. When two elements are “directly connected” indicates that one element is directly connected to another element without an intervening element.

[0023] Similar reference numerals refer to similar devices throughout the specification. Even though a reference numeral is not mentioned with reference to a drawing, the reference numeral is mentioned with reference to another drawing. A reference numeral may be shown in one or more drawings.

[0024] The present disclosure is described in more detail through examples. These examples are provided only for convenience in illustrating and describing the present disclosure, and the scope of the present disclosure is not limited by these examples.

[0025] FIG. **1** illustrates a configuration of a multichip package **10** according to an embodiment of the present disclosure.

[0026] As shown in FIG. **1**, the multichip package **10** includes a substrate **100**, a first chip **110-1**, a second chip **110-2**, a third chip **110-3**, a fourth chip **110-4**, and a support structure **120**.

[0027] The substrate **100** is supplied with a voltage through any of the solder balls **101-1** through **101-7**, for example, a voltage from a source external to the multichip package **10**. The substrate **100** includes a connection structure **103** connected to a solder ball **101-8**. The substrate **100** may be connected to a ground voltage, for example, through the solder ball **101-8**. One or more of the solder balls provides an external electrical connection for the substrate **100**, for example, to a voltage, a ground voltage, or device external to the multichip package **10**. The substrate **100** may be connected to devices external to the multichip package **10** such as a cooler or heat sink, an electrostatic discharge (ESD) protection device, and the like.

[0028] The first chip **110-1** is stacked over or on the substrate **100**. The first chip **110-1** is electrically connected to the substrate **100** through bonding wire **131-1**. The second chip **110-2** is stacked over or on the first chip **110-1**. An overhang occurs because the second chip **110-2** is offset from the first chip **110-1** in an X direction, where a portion of the second chip **110-2** is not supported by the first chip **110-1** in a Z direction. The second chip **110-2** is electrically connected to the first chip **110-1** through bonding wire **131-3**. The third chip **110-3** is stacked over or on the second chip **110-2**. An overhang occurs because the third chip **110-3** is offset from the second chip **110-2** in the X direction, where a portion of the third chip **110-3** is not supported by the second chip **110-2** in the Z direction. The third chip **110-3** is electrically connected to the second chip **110-2** through bonding wire **131-5**. The fourth chip **110-4** is stacked over or on the third chip **110-3**. An overhang occurs because the fourth chip **110-4** is offset from the third chip **110-3** in the X direction, where a portion of the fourth chip **110-4** is not supported by the third chip **110-3** in the Z direction. The fourth chip **110-4** is electrically connected to the third chip **110-3** through bonding wire **131-7**. The fourth chip **110-4** is electrically connected to the substrate **100** through bonding wire **131-8**.

[0029] The chips **110-1**, **110-2**, **110-3**, and **110-4** are stacked in the Z direction. An overhang of the second chip **110-2**, an overhang of the third chip **110-3**, and an overhang of the fourth chip result from offset stacking of the first chip **110-1**, the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4**.

[0030] The support structure **120** is disposed or connected between the substrate **100** and the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** to support the overhang of each of the second chip **110-2**, the third chip **110-3**, the fourth chip **110-4**. The support structure **120** supports the overhang where the second chip **110-2** is not supported by the first chip **110-1** in a region between the second chip **110-2** and the substrate **100**. The support structure **120** supports the overhang where the third chip **110-3** is not supported by the second chip **110-2** in a region between the third chip **110-3** and the substrate **100**. The support structure **120** supports the overhang where the fourth chip **110-4** is not supported by the third chip **110-3** in a region between the fourth chip **110-4** and the substrate **100**. The support structure **120** supports the overhang of each of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4**, such that physical deformation of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** due to the overhang of the second chip **110-2**, the overhang of the third chip **110-3**, and the overhang of the fourth chip **110-4** may be reduced or prevented.

[0031] The support structure **120** is connected to the solder ball **101-8** through the connection structure **103**. The support structure **120** and the connection structure **103** may be advantageously formed of a highly conductive material that conducts heat and electricity well. Each of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** may be connected to a ground voltage to be grounded through the support structure **120**, the connection structure **103**, and the solder ball **101-8**. Each of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** may be connected to a cooler or a heat sink through the support structure **120**, the connection structure **103**, and the solder ball **101-8**, such that heat generated by each of the first chip **110-1**, the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** is dissipated outside the multichip package **10**.

[0032] FIG. **2** and FIG. **3** illustrate top and bottom views, respectively, of a plurality of chips and a support structure **120**, for example, as shown in the multichip package **10** of FIG. **1**. FIG. **2** shows a top view in the Z direction of the plurality of stacked chips **110-1**, **110-2**, **110-3**, and **110-4** and the support structure **120**. FIG. **3** shows a bottom view in the Z direction of the plurality of stacked chips **110-1**, **110-2**, **110-3**, and **110-4** and the support structure **120**.

[0033] Referring to FIG. **2** and FIG. **3**, a plurality of connection structures **105** among the first chip **110-1**, second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** are illustrated. As shown in FIG. **2**, a section of the support structure **120** extends beyond the fourth chip **110-4** in the X direction and is connected to the solder ball **101-8** through the connection structure **103** of the substrate **100** as shown in FIG. **1**. As shown in FIG. **3**, the support structure **120** supports the overhang of each of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4**. A section of the support structure **120** extends beyond the fourth chip **110-4** in the X direction and is connected to the solder ball **101-8** through the connection structure **103** of the substrate **100** as shown in FIG. **1**. The support structure **120** supports the overhang of the second chip **110-2**, the overhang of the third chip **110-3**, and the overhang of the fourth chip **110-4** to prevent or reduce physical deformation of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4**, and may be grounded or may be connected to a cooler or a heat sink through the solder ball **101-8** to dissipate the heat generated by the first chip **110-1**, second chip **110-2**, third chip **110-3**, and fourth chip **110-4** outside the multichip package **10**.

[0034] FIG. **4** and FIG. **5** illustrate top and bottom views, respectively, of a plurality of chips and a plurality of support structures **120**, for example, as shown in FIG. **1**. FIG. **4** shows a top view in the Z direction of a plurality of stacked chips **110-1**A, **110-2**A, **110-3**A, and **110-4**A and a plurality of support structures **120** in this example. FIG. **3** shows a bottom view in the Z direction of the plurality of stacked chips **110-1**A, **110-2**A, **110-3**A, and **110-4**A and the plurality of support structures **120** in this example.

[0035] Referring to FIG. **4** and FIG. **5**, a plurality of connection structures **105** among the first chip **110-1**A, the second chip **110-2**A, the third chip **110-3**A, and the fourth chip **110-4**A are illustrated. As shown in FIG. **4**, a section of the each of the support structures **120** extends beyond the fourth chip **110-4**B in the X direction and is connected to the solder ball **101-8** through the connection structure **103** of the substrate **100** as shown in FIG. **1**. According to this example, each of the support structures **120** is connected to a separate solder ball. As shown in FIG. **5**, each of the support structures **120** supports the overhang of each of the second chip **110-2**A, the third chip **110-3**A, and the fourth chip **110-4**A. A section of each of the support structures **120** extends beyond the fourth chip **110-4**A in the X direction and is connected to the solder ball **101-8** through the connection structure **103** of the substrate **100** as shown in FIG. **1**. Each of the support structures **120** supports the overhang of the second chip **110-2**A, the overhang of the third chip **110-3**B, and the overhang of the fourth chip **110-4**B to prevent or reduce physical deformation of the second chip **110-2**A, the third chip **110-3**A, and the fourth chip **110-4**A, and may be grounded or may be connected to a cooler or a heat sink through the solder ball **101-8** to dissipate heat generated by the first chip **110-1**A, the second chip **110-2**A, the third chip **110-3**A, and the fourth chip **110-4**A outside the multichip package **10**. Optionally, the two support structures **120** may be different shapes, including different widths or lengths. More than two support structures **120** may be utilized. Bridges or connections between the support structures **120** may be provided in the Y direction, which bridges or connections may electrically and/or thermally connect the support structures **120**.

[0036] FIG. **6** illustrates a configuration of a multichip package **20** according to an embodiment of the present disclosure.

[0037] As shown in FIG. **6**, the multichip package **20** includes a substrate **200**, a first chip **210-1**, a second chip **210-2**, a third chip **210-3**, a fourth chip **210-4**, and a support structure **220**.

[0038] The substrate **200** is supplied with a voltage through any of the solder balls **201-1** through **201-7**, for example, a voltage from a source external to the multichip package **20**. The substrate **200** includes a connection structure **203** connected to a solder ball **210-8**. The substrate **200** may be connected to the ground voltage through the solder ball **201-8**. One or more of the solder balls provides an external electrical connection for the substrate **200**, for example, to a voltage, a ground voltage, or device external to the multichip package **20**. The substrate **200** may be connected to devices external to the multichip package **20** such as a cooler or a heat sink, an electrostatic discharge protection device, and the like.

[0039] The first chip **210-1** is stacked over or on the substrate **200**. The first chip **210-1** is electrically connected to the substrate **200** through bonding wire **231-1**. The second chip **210-2** is stacked over or on the first chip **210-1**. An overhang occurs because the second chip **210-2** is offset from the first chip **210-1** in the X direction, where a portion of the second chip **210-2** is not supported by the first chip **210-1** in the Z direction. The second chip **210-2** is electrically connected to the first chip **210-1** through bonding wire **231-3**. The third chip **210-3** is stacked over or on the second chip **210-2**. An overhang occurs because the third chip **210-3** is offset from the second chip **210-2** in the X direction, where a portion of the third chip **210-3** is not supported by the second chip **210-2** in the Z direction. The third chip **210-3** is electrically connected to the second chip **210-2** through bonding wire **231-5**. The fourth chip **210-4** is stacked over or on the third chip **210-3**. An overhang occurs because the fourth chip **210-4** is offset from the third chip **210-3** in the X direction, where a portion of the fourth chip **210-4** is not supported by the third chip **210-3** in the Z direction. The fourth chip **210-4** is electrically connected to the third chip **210-3** through bonding wires **231-7**. The fourth chip **210-4** is electrically connected to the substrate **200** through bonding wire **231-8**. The chips **210-1**, **210-2**, **210-3**, and **210-4** are stacked in the Z direction.

[0040] The support structure **220** is disposed or connected between the third chip **210-3** and the substrate **200** to support the overhang of the third chip **210-3**. The support structure **220** supports the overhang occurs where the third chip **210-3** is not supported by the second chip **210-2** in a region between the third chip **210-3** and the substrate **200**. The support structure **220** supports the overhang of the third chip **210-3**, such that physical deformation of at least the third chip **210-3** due to the overhang of the third chip **210-3** may be reduced or prevented.

[0041] The support structure **220** is connected to the solder ball **201-8** through the connection structure **203** of the substrate **200**. The support structure **220** and the connection structure **203** may be advantageously made of a highly conductive material that conducts heat and electricity well. The third chip **210-3** may be connected to the ground voltage to be grounded through the support structure **220**, the connection structure **203**, and the solder ball **201-8**. The third chip **210-3** may be connected to a cooler or a heat sink through the support structure **220**, the connection structure **203**, and the solder ball **201-8**, such that heat generated by at least the third chip **210-3** is dissipated outside the multichip package **20**.

[0042] FIG. **7** illustrates a configuration of a multichip package **30** according to an embodiment of the present disclosure.

[0043] As shown in FIG. **7**, the multichip package **30** includes a substrate **300**, a first chip **310-1**, a second chip **310-2**, a third chip **310-3**, a fourth chip **310-4**, a fifth chip **310-5**, a sixth chip **310-6**, a seventh chip **310-7**, an eighth chip **310-8**, and a support structure **320**.

[0044] The substrate **300** is supplied with a voltage through any of solder balls **301-1** through **301-9**, for example, a voltage from a source external to the multichip package **30**. The substrate **300** includes a connection structure **303** connected to a solder ball **301-10**. The substrate **300** may be connected to the ground voltage through the solder ball **301-10**. One or more of the solder balls provides an external electrical connection for the substrate **300**, for example, to a voltage, a ground voltage, or device external to the multichip package **30**. The substrate **300** may be connected to devices external to the multichip package **30** such as a cooler or heat sink, an electrostatic discharge protection device, and the like, for example, through the solder ball **301-10**.

[0045] The first chip **310-1** is stacked over or on the substrate **300**. The first chip **310-1** is electrically connected to the substrate **300** through bonding wire **331-1**. The second chip **310-2** is stacked over or on the first chip **310-1**. An overhang occurs because the second chip **310-2** is offset from the first chip **310-1** in an X direction, where a portion of the second chip **310-2** is not supported by the first chip **310-1** in the Z direction. The second chip **310-2** is electrically connected to the first chip **310-1** through bonding wire **331-3**. The third chip **310-3** is stacked over or on the second chip **310-2**. An overhang occurs because the third chip **310-3** is offset from the second chip **310-2** in the X direction, where a portion of the third chip **310-3** is not supported by the second chip **310-2** in the Z direction. The third chip **310-3** is electrically connected to the second chip **310-2** through bonding wire **331-5**. The fourth chip **310-4** is stacked over or on the third chip **310-3**. An overhang occurs because the fourth chip **310-4** is offset from the third chip **310-3** in the X direction, where a portion of the fourth chip **310-4** is not supported by the third chip **310-3** in the Z direction. The fourth chip **310-4** is electrically connected to the third chip **310-3** through bonding wire **331-7**. The fifth chip **310-5** is stacked over or on the fourth chip **310-4**. An overhang occurs because the fifth chip **310-5** is offset from the fourth chip **310-4** in the X direction, where a portion of the fifth chip **310-5** is not supported by the fourth chip **310-4** in the Z direction. The fifth chip **310-5** is electrically connected to the fourth chip **310-4** through bonding wire **331-9**. The sixth chip **310-6** is stacked over or on the fifth chip **310-5**. An overhang occurs because the sixth chip **310-6** is offset from the fifth chip **310-5** in the X direction, where a portion of the sixth chip **310-6** is not supported by the fifth chip **310-5** in the Z direction. The sixth chip **310-6** is electrically connected to the fifth chip **310-5** through bonding wire **331-11**. The seventh chip **310-7** is stacked over or on the sixth chip **310-6**. An overhang occurs because the seventh chip **310-7** is offset from the sixth chip **310-6** in the X direction, where in which a portion of the seventh chip **310-7** is not supported by the sixth chip **310-6** in the Z direction. The seventh chip **310-7** is electrically connected to the sixth chip **310-6** through bonding wire **331-13**. The eighth chip **310-8** is stacked over or on the seventh chip **310-7**. An overhang occurs because the eighth chip **310-8** is offset from the seventh chip **310-7** in the X direction, where a portion of the eighth chip **310-8** is not supported by the seventh chip **310-7** in the Z direction. The eighth chip **310-8** is electrically connected to the seventh chip **310-7** through bonding wire **331-15**. The eighth chip **310-8** is electrically connected to the substrate **300** through bonding wire **331-16**. The chips **310-1**, **310-2**, **310-3**, and **310-4**, **310-5**, **310-6**, **310-7**, and **310-8** are stacked in the Z direction.

[0046] The support structure **320** is disposed or connected between the substrate **300** and the third chip **310-3**, the fifth chip **310-5**, and the seventh chip **310-7** to support the overhang of each of the third chip **310-3**, the fifth chip **310-5**, the seventh chip **310-7**. The support structure **320** supports the overhang where the third chip **310-3** is not supported by the second chip **310-2** in a region between the third chip **310-3** and the substrate **300**. The support structure **320** supports the overhang where the fifth chip **310-5** is not supported by the fourth chip **310-4** in a region between the fifth chip **310-5** and the substrate **300**. The support structure **320** supports the overhang where the seventh chip **310-7** is not supported by the sixth chip **310-6** in a region between the seventh chip **310-7** and the substrate **300**. The support structure **320** supports the overhang of each of the third chip **310-3**, the fifth chip **310-5**, and the seventh chip **310-7**, such that physical deformation of at least the third chip **310-3**, the fifth chip **310-5**, and the seventh chip **310-7** due to the overhang of the third chip **310-3**, the overhang of the fifth chip **310-5**, and the overhang of the seventh chip **310-7** may be reduced or prevented.

[0047] The support structure **320** is connected to the solder ball **301-10** through the connection structure **303** of the substrate **300**. The support structure **320** and the connection structure **303** may be advantageously made of a highly conductive material that conducts heat and electricity well. Each of the third chip **310-3**, the fifth chip **310-5**, and the seventh chip **310-7** may be connected to the ground voltage to be grounded through the support structure **320**, the connection structure **303**, and the solder ball **301-10**. Each of the third chip **310-3**, the fifth chip **310-5**, and the seventh chip **310-7** may be connected to a cooler or a heat sink through the support structure **320**, the connection structure **303**, and the solder ball **301-10** to dissipate heat generated by at least the third chip **310-3**, the fifth chip **310-5**, and the seventh chip **310-7** outside the multichip package **30**.

[0048] FIG. **8** illustrates a configuration of a multichip package **40** according to an embodiment of the present disclosure.

[0049] As shown in FIG. **8**, the multichip package **40** includes a substrate **400**, a first chip **410-1**, a second chip **410-2**, a third chip **410-4**, a fourth chip **410-4**, a first support structure **420-1**, a second support structure **420-2**, and a third support structure **420-3**. The substrate **400** is supplied with a voltage through any of solder balls **401-1** through **401-4** and **401-7**, for example, a voltage from a source external to the multichip package **40**. The substrate **400** includes a first connection structure **403-1** connected to a solder ball **401-5**, a second connection structure **403-2** connected to a solder ball **401-6**, and a third connection structure **403-3** connected to a solder ball **401-8**. The substrate **400** may be connected to the ground voltage, for example, through the solder balls **401-5**, **401-6**, **401-8**. One or more of the solder balls provides an external electrical connection for the substrate **400**, for example, to a voltage, a ground voltage, or device external to the multichip package **40**. The substrate **400** may be connected to devices external to the multichip package **40** such as a cooler or heat sink, an electrostatic discharge protection device, and the like through the solder balls **401-5**, **401-6**, **401-8**.

[0050] The first chip **410-1** is stacked over or on the substrate **400**. The first chip **410-1** is electrically connected to the substrate **400** through bonding wire **431-1**. The second chip **410-2** is stacked over or on the first chip **410-1**. An overhang occurs because the second chip **410-2** is offset from the first chip **410-1** in an X direction, where a portion of the second chip **410-2** is not supported by the first chip **410-1** in the Z direction. The second chip **410-2** is electrically connected to the first chip **410-1** through bonding wire **431-3**. The third chip **410-3** is stacked over or on the second chip **410-2**. An overhang occurs because the third chip **410-3** is offset from the second chip **410-2** in the X direction, where a portion of the third chip **410-3** is not supported by the second chip **410-2** in the Z direction. The third chip **410-3** is connected to the second chip **410-2** through bonding wire **431-5**. The fourth chip **410-4** is stacked over or on the third chip **410-3**. An overhang occurs because the fourth chip **410-4** is offset from the third chip **410-3** in the X direction, where a portion of the fourth chip **410-4** is not supported by the third chip **410-3** in the Z direction. The fourth chip **410-4** is electrically connected to the third chip **410-3** through bonding wire **431-7**. The fourth chip **410-4** is electrically connected to the substrate **400** through bonding wire **431-8**.

[0051] The chips **410-1**, **410-2**, **410-3**, and **410-4** are stacked in the Z direction.

[0052] The first support structure **420-1** supports the overhang where the second chip **410-2** is not supported by the first chip **410-1** in a region between the second chip **410-2** and the substrate **400**.

The second support structure **420-2** supports the overhang where the third chip **410-3** is not supported by the second chip **410-2** in a region between the third chip **410-3** and the substrate **400**. The third support structure **420-3** supports the overhang where the fourth chip **410-4** is not supported by the third chip **410-3** in a region between the fourth chip **410-4** and the substrate **400**. The first support structure **420-1**, the second support structure **420-2**, and the third support structure **420-3** support the overhang of the second chip **410-2**, the overhang of the third chip **420-3**, and the overhang of the fourth chip **410-4**, respectively, such that physical deformation of at least the second chip **410-2**, the third chip **410-3**, and the fourth chip **410-4** due to the overhangs of the second chip **410-2**, the third chip **410-3**, and the fourth chip **410-4** may be reduced or prevented.
[0053] The first support structure **420-1**, the second support structure **420-2**, and the third support structure **420-1** through **420-3** are connected to the solder balls **401-5**, **401-6**, and **401-8** through the first connection structure **403-1**, the second connection structure **403-2**, and the third connection structure **403-3**, respectively. Each of the support structures **420-1** through **420-3** and each of the connection structures **403-1** through **403-3** may be advantageously formed of a highly conductive material that conducts heat and electricity well. The second chip **410-2**, the third chip **410-3**, and the fourth chip **410-4** may be connected to the ground voltage to be grounded through the support structures **420-1** through **420-3**, the connection structures **403-1** through **403-3**, and the solder balls **401-5**, **401-6**, and **401-8**, respectively. The second chip **410-2**, the third chip **410-3**, and the fourth chip **410-4** may be connected to a cooler or a heat sink through the support structures **420-1** through **420-3**, the connection structures **403-1** through **403-3**, and the solder balls **401-5**, **401-6**, and **401-8**, respectively, to dissipate heat generated by the first chip **410-1**, the second chip **410-2**, the third chip **410-3**, and the fourth chip **410-4** outside the multichip package **40**.
[0054] FIG. **9** illustrates a configuration of a multichip package **50** according to an embodiment of the present disclosure.
[0055] As shown in FIG. **9**, the multichip package **50** includes a substrate **500**, a first chip **510-1**, a second chip **510-2**, a third chip **510-3**, a fourth chip **510-4**, a first support structure **520-1**, and a second support structure **520-2**.
[0056] The substrate **500** is supplied with a voltage through any of solder balls **501-1** through **501-4** and **501-7**, for example, a voltage from a source external to the multichip package **50**. The substrate **500** includes a first connection structure **503-1** connected to solder balls **501-5** and **501-6** and a second connection structure **503-2** connected to a solder ball **501-8**. One or more of the solder balls provides an external electrical connection for the substrate **500**, for example, to a voltage, a ground voltage, or device external to the multichip package **50**. The substrate **500** may be connected to the ground voltage, for example, through the solder balls **501-5**, **501-6**, and **501-8**. The substrate **500** may be connected to devices external to the multichip package **40** such as a cooler or heat sink, an electrostatic discharge (ESD) protection device, and the like.
[0057] The first chip **510-1** is stacked over or on the substrate **500**. The first chip **510-1** is electrically connected to the substrate **500** through bonding wire **531-1**. The second chip **510-2** is stacked over or on the first chip **510-1**. An overhang occurs because the second chip **510-2** is offset from the first chip **510-1** in an X direction, where a portion of the second chip **510-2** is not supported by the first chip **510-1** in the Z direction. The second chip **510-2** is electrically connected to the first chip **510-1** through bonding wire **531-3**. The third chip **510-3** is stacked over or on the second chip **510-2**. An overhang occurs because the third chip **510-3** is offset from the second chip **510-2** in the X direction, where a portion of the third chip **510-3** is not supported by the second chip **510-2** in the Z direction. The third chip **510-3** is electrically connected to the second chip **510-2** through bonding wire **531-5**. The fourth chip **510-4** is stacked over or on the third chip **510-3**. An overhang occurs because the fourth chip **510-4** is offset from the third chip **510-3** in the X direction, where a portion of the fourth chip **510-4** is not supported by the third chip **510-3** in the Z direction. The fourth chip **510-4** is electrically connected to the third chip **510-3** through bonding wire **531-7**. The fourth chip **510-4** is electrically connected to the substrate **500** through bonding

wire **531-8**. The chips **510-1**, **510-2**, **510-3**, and **510-4** are stacked in the Z direction.
[0058] The first support structure **520-1** supports the overhang of the second chip **510-2** where the second chip **510-2** is not supported by the first chip **510-1** in a region between the second chip **310-2** and the substrate **500** and supports the overhang of the third chip **510-3** where the third chip **510-3** is not supported by the second chip **510-2** in a region between the third chip **310-2** and the substrate **500**. The second support structure **520-2** supports the overhang where the fourth chip **510-4** is not supported by the third chip **510-3** in a region between the fourth chip **510-4** and the substrate **500**. The first support structure **520-1** and the second support structure **520-2** support the overhangs of the second chip **510-2**, the third chip **520-3**, and the fourth chip **510-4**, such that physical deformation of at least the second chip **510-2**, the third chip **510-3**, and the fourth chip **510-4** due to the overhangs of the second chip **510-2**, third chip **510-3**, and fourth chip **510-4** may be reduced or prevented.
[0059] The support structures **520-1** and **520-2** are connected to the solder balls **501-5**, **501-6**, and **501-8** through the connection structures **503-1** and **503-2**. Each of the support structures **520-1** and **520-2** and each of the connection structures **503-1** and **503-2** may be advantageously formed of a highly conductive material that conducts heat and electricity well. The second chip **510-2**, the third chip **610-3**, and the fourth chip **610-4** may be connected to the ground voltage to be grounded through the support structures **520-1** and **520-2**, the connection structures **503-1** and **503-2**, and the solder balls **501-5**, **501-6**, and **501-8**. The second chip **510-2**, the third chip **510-3**, and the fourth chip **510-4** may be connected to a cooler or a heat sink through the support structures **520-1** and **520-2**, the connection structures **503-1** and **503-2**, and the solder balls **501-5**, **501-6**, and **501-8**, to dissipate heat generated by the first chip **510-1**, the second chip **510-2**, the third chip **510-3**, and the fourth chip **510-4** outside the multichip package **50**.
[0060] FIG. **10** through FIG. **13** illustrate views of a multichip package formed utilizing a method of forming a multichip package **10** according to an embodiment of the present disclosure.
[0061] As shown in FIG. **10**, a substrate **100** connected to solder balls **101-1** through **101-8** is prepared or formed. One or more of the solder balls **101-1** through **101-7** provides an external electrical connection for the substrate **100**, for example, to a voltage, a ground voltage, or device external to the multichip package **10**. The substrate **100** includes a first connection structure **103** connected to the solder ball **101-8** that provides an external electrical connection for the substrate **100**, for example, to the ground voltage or devices external to the multichip package **10** such as a cooler or heat sink, an electrostatic discharge (ESD) protection device, and the like.
[0062] As shown in FIG. **11**, a support structure **120** is formed over or on the substrate **100**. The support structure **120** is electrically connected to the solder ball **101-8** through the connection structure **103**.
[0063] As shown in FIG. **12** and FIG. **13**, a first chip **110-1** is stacked over or on the substrate **100**, a second chip **110-2** is stacked over or on the first chip **110-1**, a third chip **110-3** is stacked over or on the second chip **110-2**, and a fourth chip **110-4** is stacked over or on the third chip **110-3**. As shown in FIG. **13**, the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** are stacked over or on the first chip **110-1**, the second chip **110-2**, and the third chip **110-3**, respectively, such that an overhang results where a portion of the second chip **110-2** is not supported by the first chip **110-1**, an overhang results where a portion of the third chip **110-3** is not supported by the second chip **110-2**, and an overhang results where a portion of the fourth chip **110-4** is not supported by the third chip **110-3**. The overhang of the second chip **110-2**, the overhang of the third chip **110-3**, and the overhang of the fourth chip **110-4** are supported by the support structure **120**. Because the overhangs of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** are supported by the support structure **120**, physical deformation of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** due to the overhangs of the second chip **110-2**, the third chip **110-3**, and the fourth chip **110-4** may be reduced or prevented.
[0064] The multichip package **20** shown in FIG. **6**, the multichip package **30** shown in FIG. **7**, the

multichip package **40** shown in FIG. **8**, and the multichip package **50** shown in FIG. **9** may be formed using the method of forming the multichip package **10** described with respect to FIG. **10** through FIG. **13**. Different quantities of chips and different quantities of support structures than those shown in the examples described may be utilized in any combination of chips and support structures.

[0065] Concepts are disclosed in conjunction with examples and embodiments as described above. Those skilled in the art will understand that various modifications, additions, and substitutions are possible without departing from the scope and technical spirit of the present disclosure. Accordingly, the embodiments disclosed in the present specification should be considered from an illustrative standpoint and not a restrictive standpoint. Therefore, the scope of the present disclosure is not limited to the above descriptions. All changes within the meaning and range of equivalency of the claims scope are included within their scope.

Claims

1. A multichip package comprising: a substrate connected to a plurality of solder balls including a first solder ball; a first chip stacked over the substrate; a second chip stacked over the first chip and including a first overhang not supported by the first chip; and a support structure supporting the first overhang and connected to the first solder ball that provides an external electrical connection to the substrate for a voltage external to the substrate.

2. The multichip package of claim 1, wherein the external electrical connection provides access to a ground voltage and a voltage different from the ground voltage.

3. The multichip package of claim 1, wherein the support structure comprises a plurality of support structures supporting the first overhang in at least one region between the substrate and the first overhang.

4. The multichip package of claim 1, further comprising a third chip stacked over the second chip and including a second overhang not supported by the second chip.

5. The multichip package of claim 4, wherein the support structure supports the second overhang.

6. The multichip package of claim 5, wherein the support structure comprises a plurality of support structures supporting the second overhang in at least one region between the substrate and the second overhang.

7. The multichip package of claim 4, further comprising a fourth chip stacked over the third chip and including a third overhang not supported by the third chip.

8. The multichip package of claim 7, wherein the support structure supports at least one of the second overhang and the third overhang.

9. The multichip package of claim 8, wherein the support structure comprises a plurality of support structures supporting the third overhang in at least one region between the substrate and the third overhang.

10. A multichip package comprising: a substrate connected to a plurality of solder balls including a first solder ball; a first chip stacked over the substrate; a second chip stacked over the first chip and including a first overhang not supported by the first chip; and a support structure supporting the first overhang and connected to the first solder ball that provides an external electrical connection for a device external to the substrate.

11. The multichip package of claim 10, wherein the external electrical connection provides access to a device external to the multichip package, wherein the device is a cooler, heat sink, or an electrostatic discharge (ESD) protection device.

12. The multichip package of claim 10, wherein the support structure comprises a plurality of support structures supporting the first overhang in at least one region between the substrate and the first overhang.

13. The multichip package of claim 10, further comprising a third chip stacked over the second chip and including a second overhang not supported by the second chip, wherein the support structure supports the second overhang.

14. The multichip package of claim 10, wherein the support structure comprises a plurality of support structures supporting the second overhang in at least one region between the substrate and the second overhang.

15. The multichip package of claim 10, further comprising: a third chip stacked over the second chip and including a second overhang not supported by the second chip; and a fourth chip stacked over the third chip and including a third overhang not supported by the third chip.

16. The multichip package of claim 15, wherein the support structure supports at least one of the second overhang and the third overhang.

17. The multichip package of claim 16, wherein the support structure comprises a plurality of support structures supporting the second overhang in at least one region between the substrate and the second overhang and the third overhang in at least one region between the substrate and the third overhang.

18. A multichip package comprising: a substrate connected to a plurality of solder balls including a first solder ball and a second solder ball; a first chip stacked over the substrate; a second chip stacked over the first chip and including a first overhang not supported by the first chip; a third chip stacked over the second chip and including a second overhang not supported by the second chip; a first support structure supporting the first overhang and connected to the first solder ball that provides a first external electrical connection for the substrate; and a second support structure supporting the second overhang and connected to the second solder ball that provides a second external electrical connection for the substrate.

19. A multichip package comprising: a substrate connected to a plurality of solder balls including a first solder ball and a second solder ball; a first chip stacked over the substrate; a second chip stacked over the first chip and including a first overhang not supported by the first chip; a third chip stacked over the second chip and including a second overhang not supported by the second chip; a fourth chip stacked over the third chip and including a third overhang not supported by the third chip; a first support structure supporting the first overhang and the second overhang and connected to the first solder ball that provides an external electrical connection to the substrate for a first voltage; and a second support structure supporting the third overhang and connected to the second solder ball that provides an external electrical connection to the substrate for a second voltage.

20. A method of forming a multichip package, the method comprising: preparing a substrate connected to a plurality of solder balls; forming a support structure over the substrate, wherein the support structure is connected an external electrical connection for the substrate; stacking a first chip over the substrate; and stacking a second chip over the first chip and the support structure, wherein an overhang of the second chip is not supported by the first chip and is supported by the support structure.

21. A multichip package comprising: a substrate; a first chip stacked over the substrate; a second chip stacked over the first chip, wherein the second chip is offset from the first chip in a first direction resulting in an overhang; and a support structure disposed between the overhang and the substrate and electrically connected to an external electrical connection for the substrate.